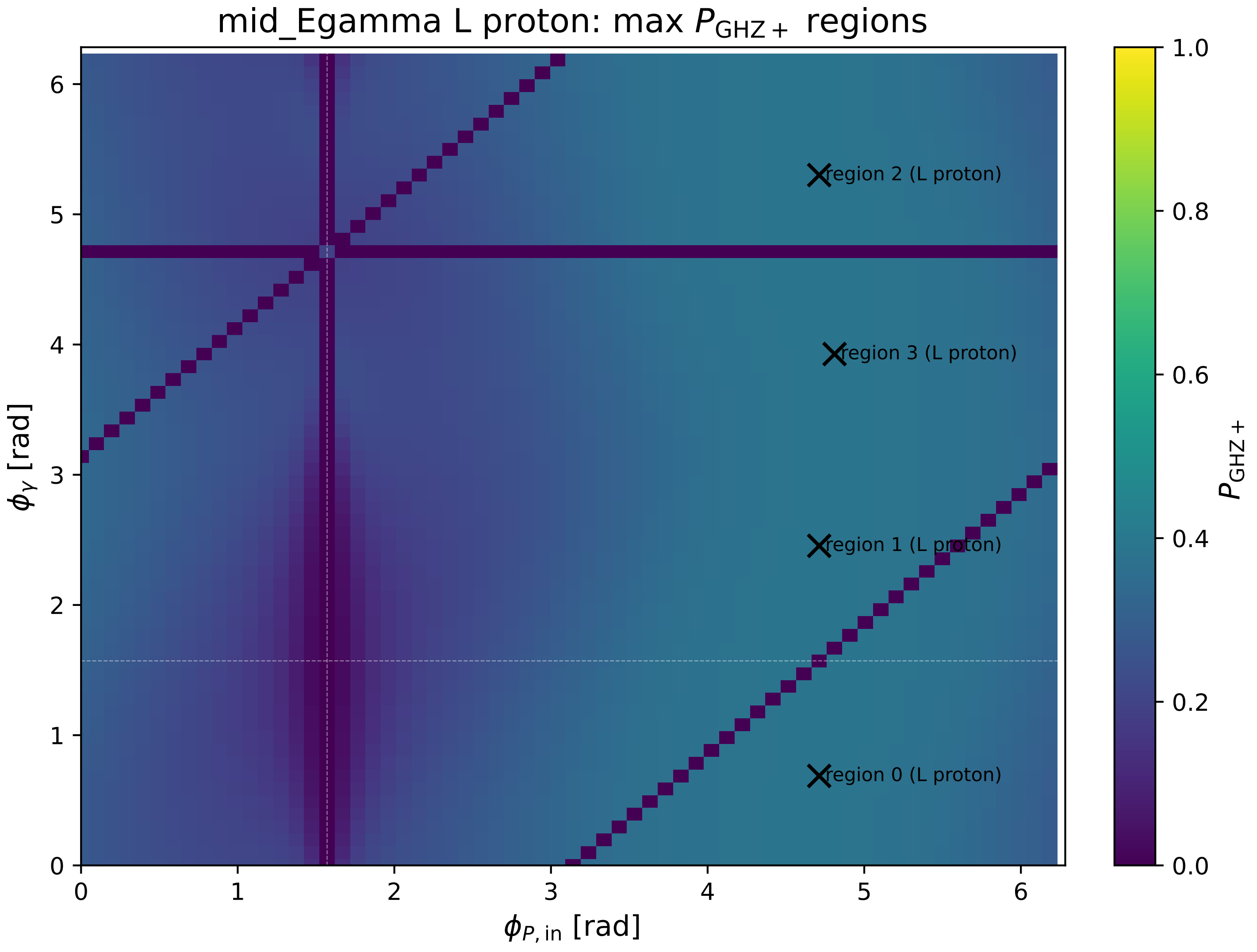
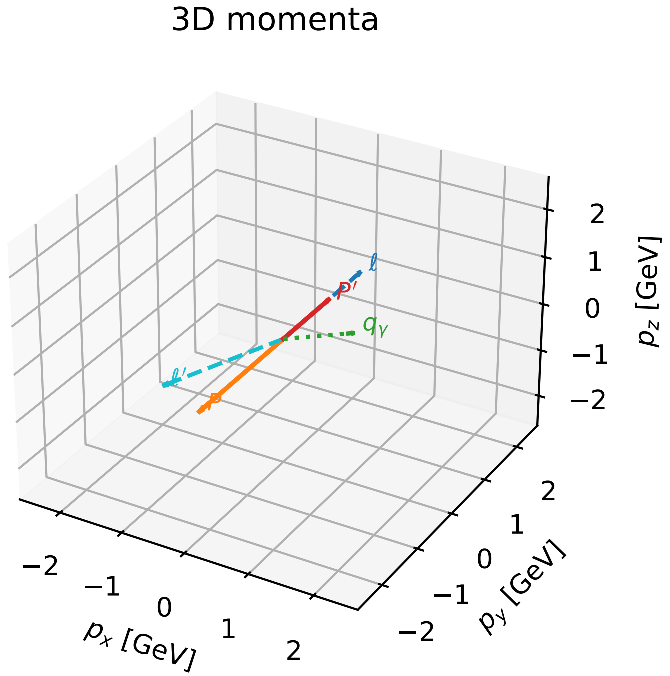


mid_Egamma L proton: max $P_{\mathrm{GHZ+}}$ regions
region 2 (L proton)
region 3 (L proton)
region 1 (L proton)
region 0 (L proton)
ϕ_{γ} [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$P_{\mathrm{GHZ+}}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

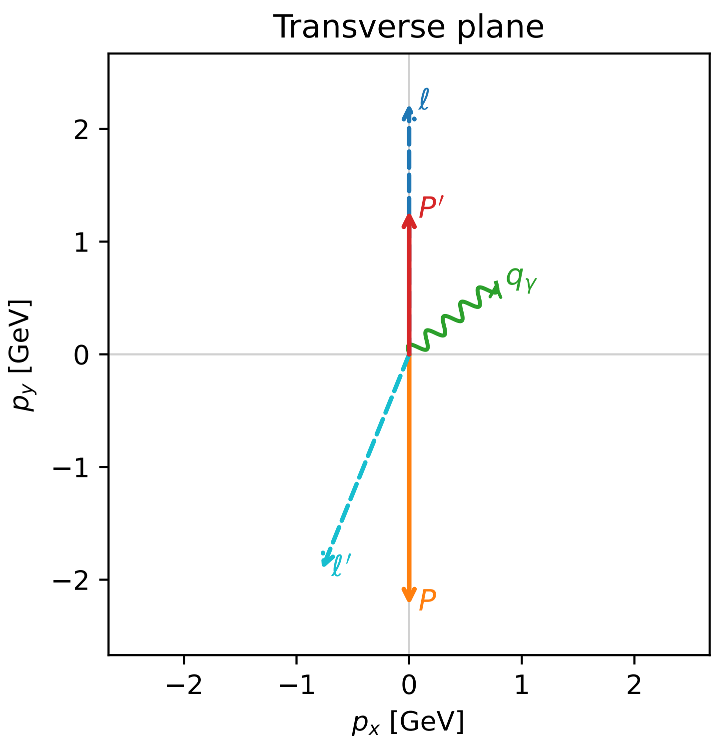
L proton characteristic max $P_{\rm GHZ+}$ configuration

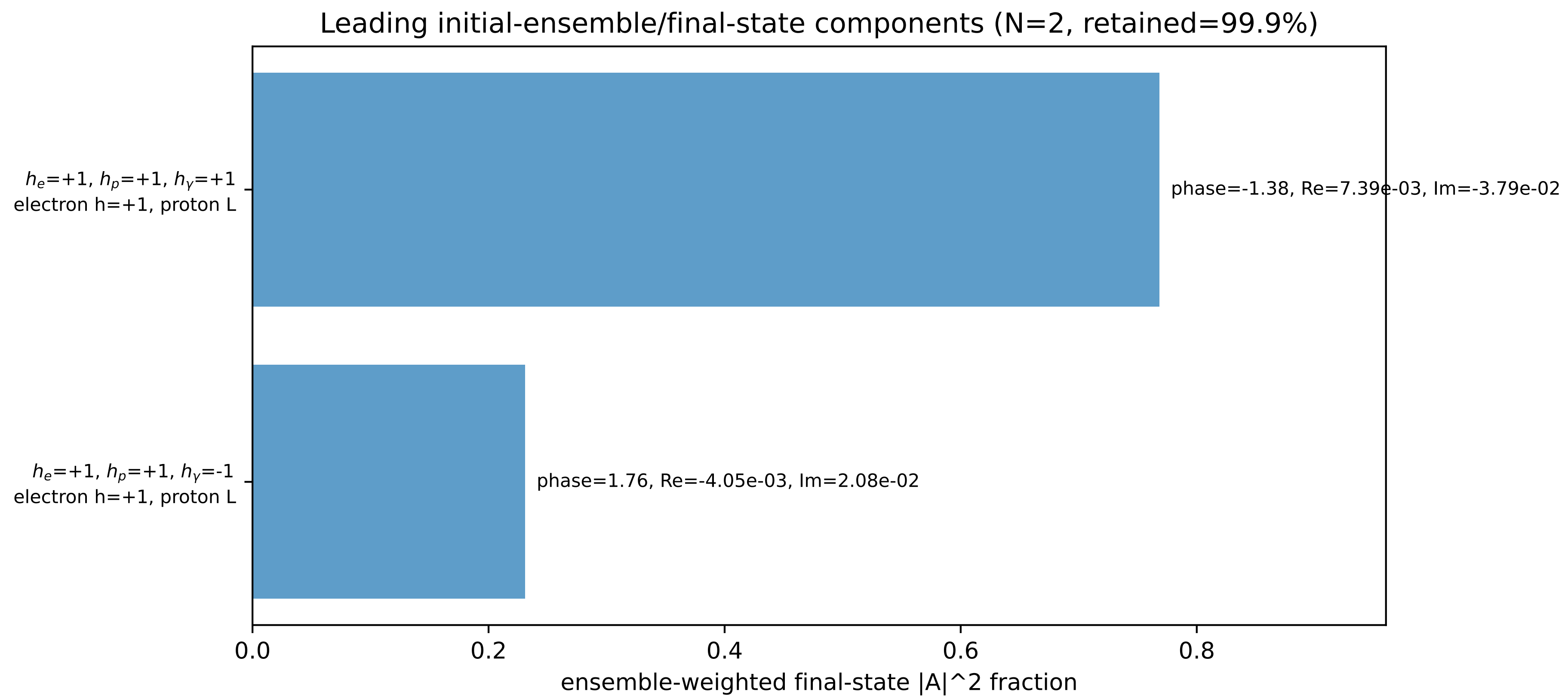


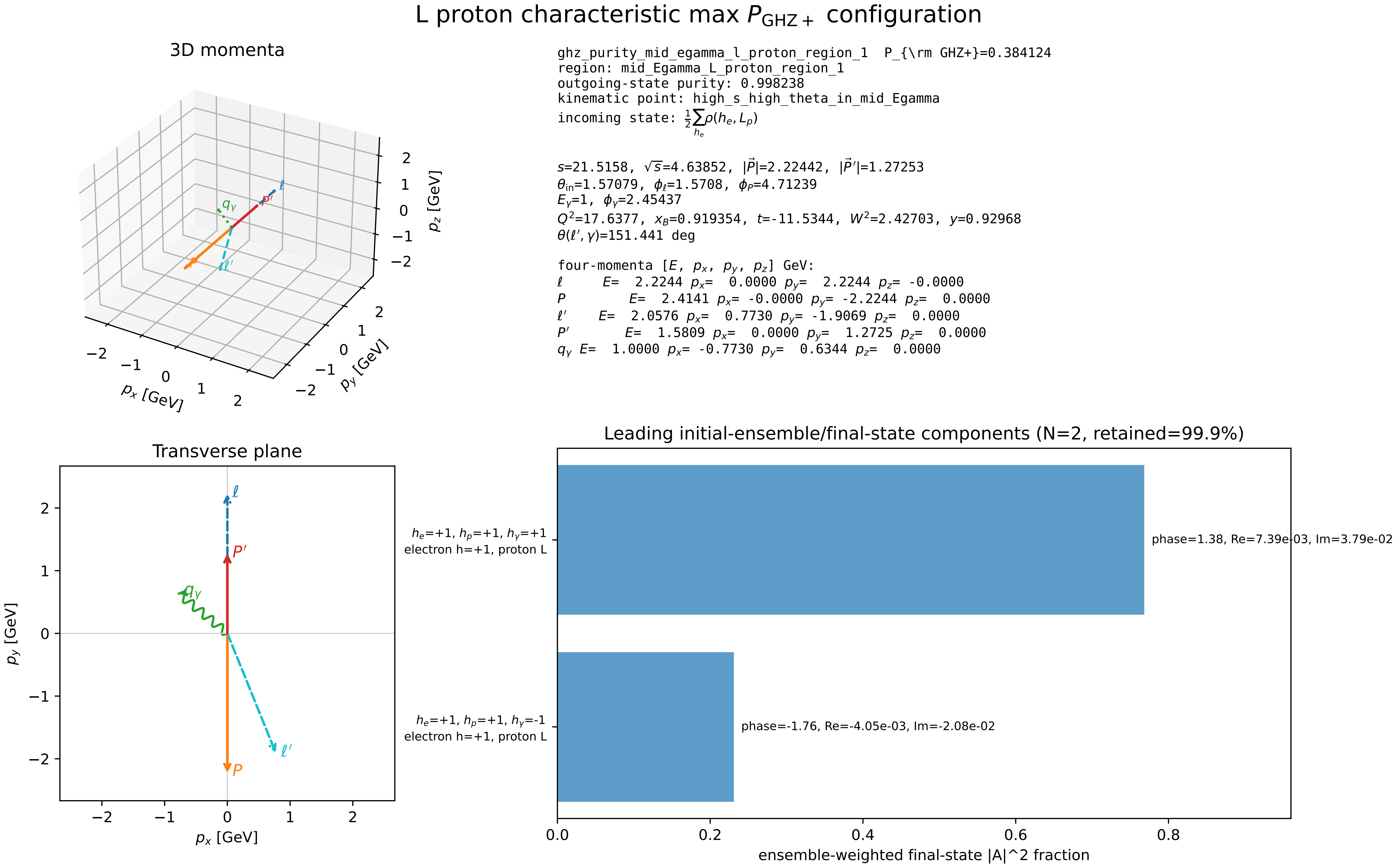
ghz_purity_mid_egamma_l_proton_region_0 P_{\rm GHZ+}=0.384124
region: mid_Egamma_L_proton_region_0
outgoing-state purity: 0.998238
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.27253
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=0.687223
Q^2=17.6377, x_B=0.919354, t=-11.5344, W^2=2.42703, y=0.92968
$\theta(\ell', \gamma)$=151.441 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 2.0576 p_x= -0.7730 p_y= -1.9069 p_z= 0.0000
P' E= 1.5809 p_x= 0.0000 p_y= 1.2725 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.7730 p_y= 0.6344 p_z= 0.0000



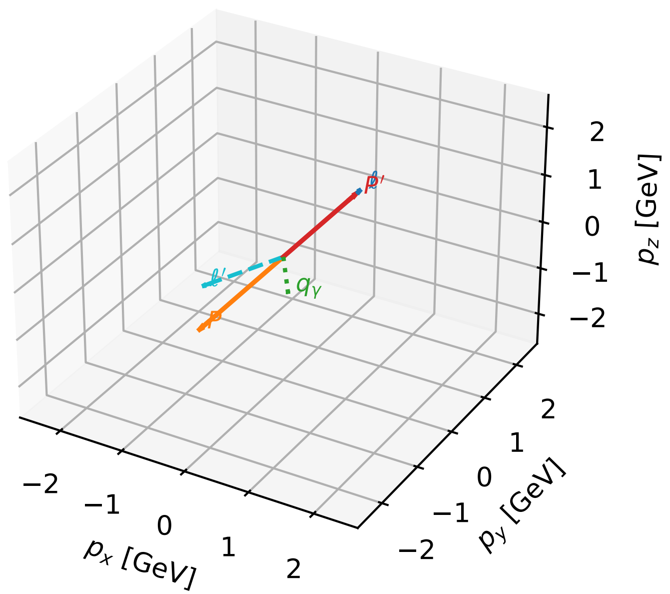




L proton characteristic max $P_{\rm GHZ+}$ configuration
3D momenta
ℓ
P'
q_γ
P
ℓ'
p_x [GeV]
p_y [GeV]
p_z [GeV]
ghz_purity_mid_egamma_l_proton_region_1 P_{\rm GHZ+}=0.384124
region: mid_Egamma_L_proton_region_1
outgoing-state purity: 0.998238
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.27253
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=2.45437
Q^2=17.6377, x_B=0.919354, t=-11.5344, W^2=2.42703, y=0.92968
$\theta(\ell',\gamma)$=151.441 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 2.0576 p_x= 0.7730 p_y= -1.9069 p_z= 0.0000
P' E= 1.5809 p_x= 0.0000 p_y= 1.2725 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.7730 p_y= 0.6344 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=2, retained=99.9%)
h_e=+1, h_p=+1, h_γ=+1
electron h=+1, proton L
phase=1.38, Re=7.39e-03, Im=3.79e-02
h_e=+1, h_p=+1, h_γ=-1
electron h=+1, proton L
phase=-1.76, Re=-4.05e-03, Im=-2.08e-02
ensemble-weighted final-state |A|^2 fraction

L proton characteristic max $P_{\mathrm{GHZ}+}$ configuration

3D momenta

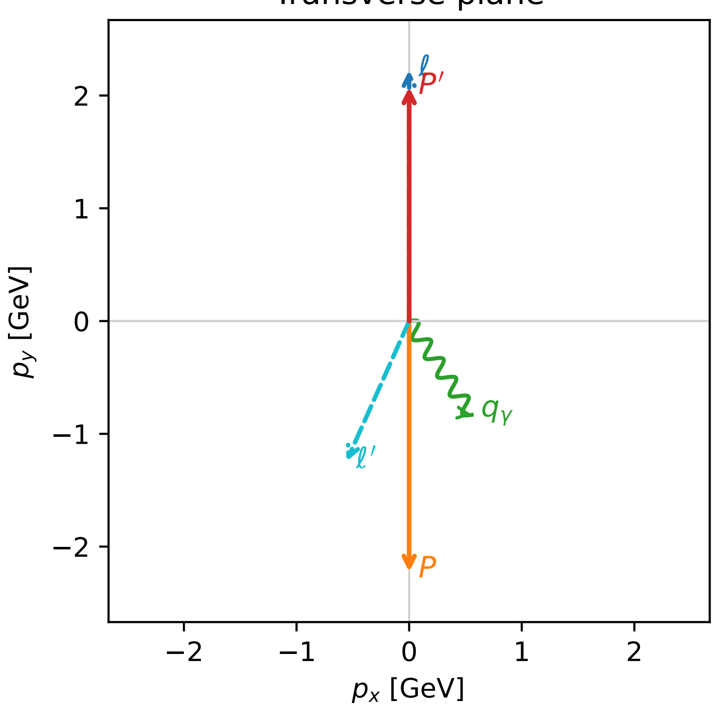


ghz_purity_mid_egamma_l_proton_region_2 P_{\rm GHZ+}=0.383883
region: mid_Egamma_L_proton_region_2
outgoing-state purity: 0.998875
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$

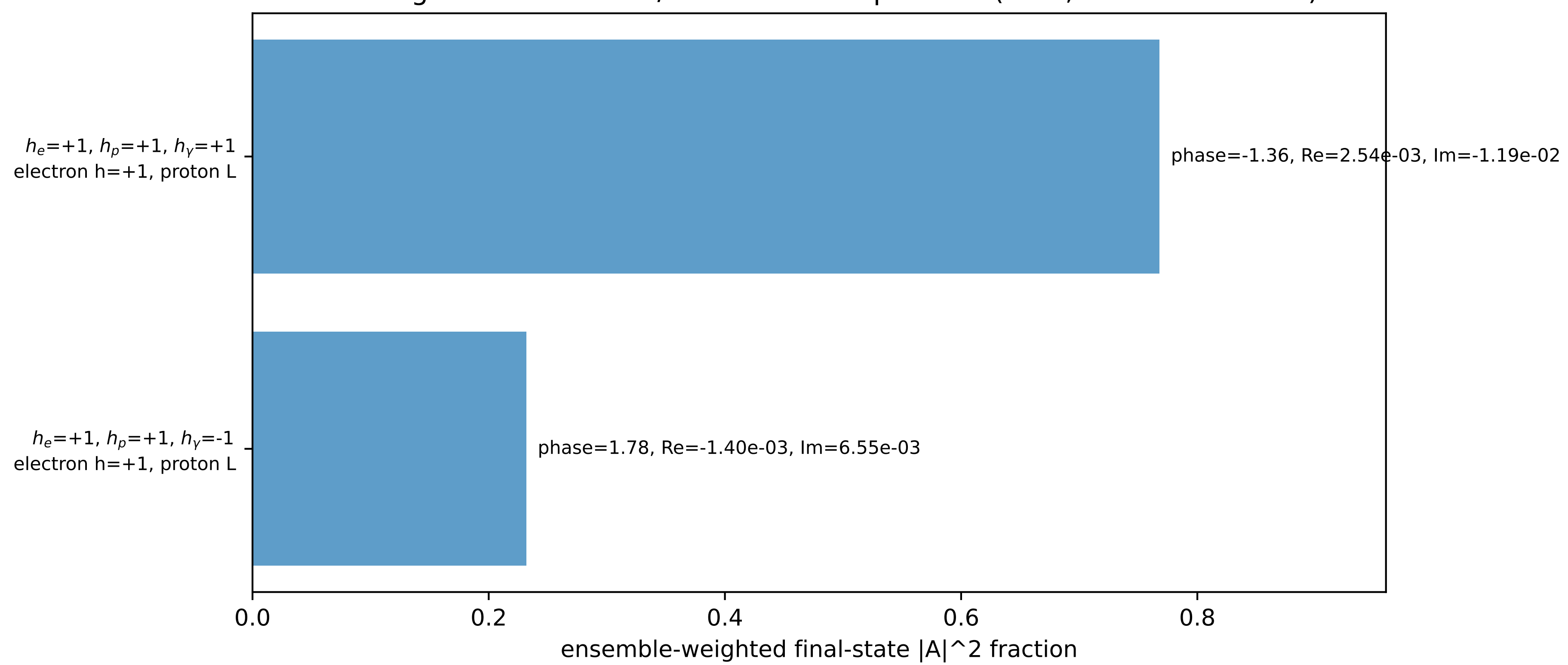
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.07465
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=5.30144
Q^2=11.5886, x_B=0.591487, t=-18.4631, W^2=8.88355, y=0.949422
$\theta(\ell', \gamma)$=57.8296 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 1.3617 p_x= -0.5556 p_y= -1.2432 p_z= 0.0000
P' E= 2.2768 p_x= 0.0000 p_y= 2.0746 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.5556 p_y= -0.8315 p_z= 0.0000

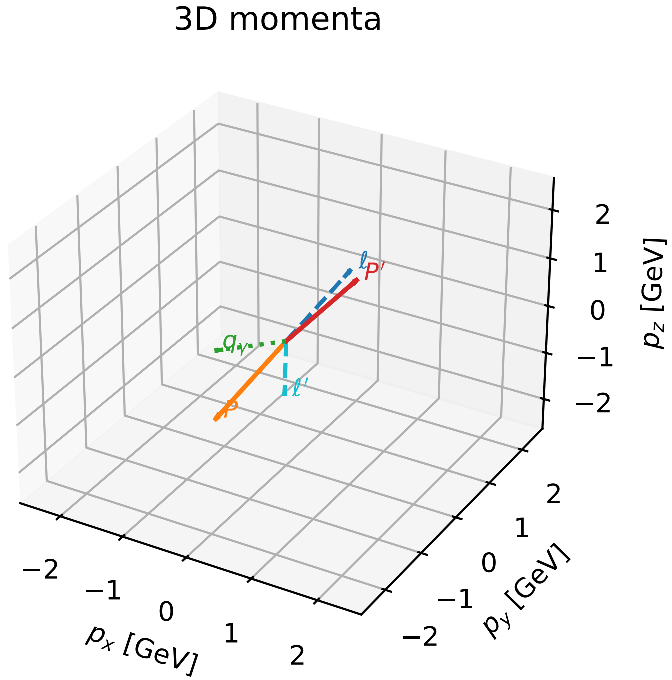
Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=99.9%)



L proton characteristic max $P_{\mathrm{GHZ}+}$ configuration




```
ghz_purity_mid_egamma_l_proton_region_3  P_{\rm GHZ+}=0.383828
region: mid_Egamma_L_proton_region_3
outgoing-state purity: 0.998632
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.9752
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.66897, ϕ_P=4.81056
E_γ=1, ϕ_γ=3.92699
Q^2=12.382, x_B=0.633397, t=-17.5427, W^2=8.0464, y=0.947304
$\theta(\ell', \gamma)$=74.1447 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.2180 p_y= 2.2137 p_z= -0.0000
P E= 2.4141 p_x= 0.2180 p_y= -2.2137 p_z= 0.0000
ℓ' E= 1.4519 p_x= 0.7071 p_y= -1.2681 p_z= 0.0000
P' E= 2.1866 p_x= 0.0000 p_y= 1.9752 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.7071 p_y= -0.7071 p_z= 0.0000